IR: MCP4725 0x60 -> LM358 -> 2N4401 -> LED

RED: MCP4725 0x61 -> LM358 -> 2N4401 -> LED

J1

MCP4725 IR 0x60

J2

PI I2C INPUT / IR OUT

DAC_IR

GND

SCL

SDA

3V3

GND

Module ADDR: GND -> 0x60

Both GND pins share 0 V. OUT is a separate signal.

J2: OUT/GND = 2.54 mm; SCL/SDA/3V3/GND = 2.00 mm

J3

MCP4725 RED 0x61

J4

RED DAC OUT / GND

DAC_RED

GND

SCL

SDA

3V3

GND

Module ADDR: VCC -> 0x61

Both GND pins share 0 V. OUT is a separate signal.

5V

J5

IR LED A / K

LED_K_IR

R1

10k 1%

CMD_IR

U1A

LM358P / DIP8 socket

R3

1k

AMP_IR

BASE_IR

Q1

2N4401 / E-B-C

R2

10k 1%

SENSE_IR

R4

82R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 82 ohm)

5V

J6

RED LED A / K

LED_K_RED

R6

10k 1%

CMD_RED

U1B

LM358P / DIP8 socket

R8

1k

AMP_RED

BASE_RED

Q2

2N4401 / E-B-C

R7

10k 1%

SENSE_RED

R9

100R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 100 ohm)

C2

DNP loop comp

AMP_IR

SENSE_IR

C2: DNP – DO NOT FIT until loop measured

C4

DNP loop comp

AMP_RED

SENSE_RED

C4: DNP – DO NOT FIT until loop measured

SHARED SUPPLY: U1A + U1B + U1C = ONE LM358 DIP-8

J7

5V INPUT / GND

5V

GND

C7

10uF 16V

C5

100nF X7R 50V

U1C

LM358P / DIP8 socket

V+

V-

C5: ceramic, non-polar. C7: electrolytic, + to 5V.

MCP4725 VCC = 3.3 V ONLY; J2 is the only Pi cable input.

ADDR is configured on each module, not by this carrier PCB.

H1

M3 / 3.2mm

H2

M3 / 3.2mm

H3

M3 / 3.2mm

H4

M3 / 3.2mm

J2 powers and controls both DAC modules over one shared PI I2C bus.

OUT pins are DAC outputs: never connect them together.

Matching labels connect SCL, SDA, 3V3 and GND between modules.

For RED: remove the old ADDR-GND bridge before selecting VCC.

Nguyen Nhat Huy – Pham Thanh Vy

Sheet: /

File: led_ir_driver.kicad_sch

Title: DATN: PPG–Simulator – LED / IR driver

Size: A3 | Date: | **Rev: 1.6**

KiCad E.D.A. 10.0.6 | Id: 1/1